

$$(40) I_n = \int x^n \cdot e^{-x} dx$$

$$\text{Let } u = x^n \Rightarrow du = n \cdot x^{n-1} dx \quad dv = e^{-x} dx \quad v = -e^{-x}$$

$$I_n = \int x^n \cdot e^{-x} dx = \int u dv = uv - \int v du = x^n \cdot (-e^{-x}) - \int (-e^{-x}) n \cdot x^{n-1} dx$$

$$= -e^{-x} \cdot x^n + n \int e^{-x} x^{n-1} dx = n \cdot I_{n-1} - e^{-x} x^n$$

$$I_n = -e^{-x} x^n + n \cdot I_{n-1}$$

$$I_0 = -e^{-x} + C_1$$

$$I_1 = -x \cdot e^{-x} + I_0 = -x e^{-x} - e^{-x} + C_2$$

$$I_2 = -x^2 e^{-x} + 2I_1 =$$

⋮

$$(41) I_n = \int \cos^n x dx = \int \cos^{n-1} x \cos x dx$$

$$\cos^{n-1} x = u \Rightarrow du = (n-1) \cos^{n-2} x (-\sin x) dx$$

$$\cos x dx = dv \Rightarrow v = \sin x$$

$$\int \cos^{n-1} x \cos x dx = \int u dv = u \cdot v - \int v du = \cos^{n-1} x \cdot \sin x - \int \sin x (n-1) \cos^{n-2} x (-\sin x) dx$$

$$\int \cos^n x dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x \sin^2 x dx \quad \sin^2 x = 1 - \cos^2 x$$

$$= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx - (n-1) \int \cos^n x dx$$

$$n \int \cos^n x dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx$$

$$\Rightarrow n I_n = \cos^{n-1} x \sin x + (n-1) I_{n-2}$$

$$I_n = \frac{1}{n} \cos^{n-1} x \sin x + \frac{n-1}{n} I_{n-2}$$

$$④ I_n = \int \sin^n x dx = \int \sin^{n-1} x \cdot \sin x dx$$

$$\sin^{n-1} x = u \Rightarrow (n-1) \sin^{n-2} x \cos x dx = du, \quad \sin x dx = dv \Rightarrow v = -\cos x$$

$$I_n = \int \sin^n x dx = \int u \cdot dv = u \cdot v - \int v \cdot du = \sin^{n-1} x \cdot (-\cos x) + (n-1) \int \sin^{n-2} x \cos^2 x dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx$$

$$n \int \sin^n x dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x dx$$

$$\Rightarrow I_n = -\frac{1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}$$

$$④ I_n = \int \frac{dx}{(x^2 + a^2)^n}$$

$$x = a \cdot \tan t \Rightarrow dx = \frac{a dt}{\cos^2 t}$$

$$(x^2 + a^2)^n = (a^2 \tan^2 t + a^2)^n = \left(\frac{a^2}{\cos^2 t} \right)^n = \frac{a^{2n}}{\cos^{2n} t}$$

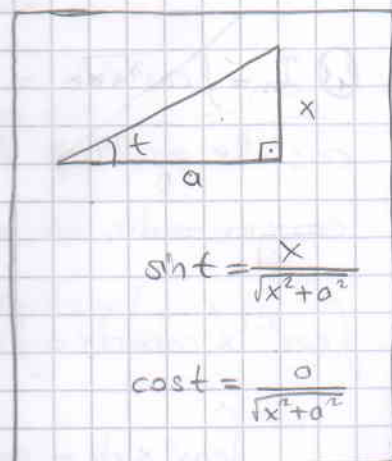
$$I_n = \int \frac{\frac{a \cdot dt}{\cos^2 t}}{\frac{a^{2n}}{\cos^{2n} t}} = \frac{1}{a^{2n-1}} \int \cos^{2n-2} t dt$$

$$I_n = \frac{1}{a^{2n-1}} \int \cos^{2n-2} t \cdot dt = \frac{1}{a^{2n-1}} \left[\frac{1}{2n-2} \cos^{2n-3} t \sin t + \frac{2n-3}{2n-2} \int \cos^{2n-4} t dt \right]$$

$$\int \cos^{2n-2} t dt = a^{2n-1} I_n \Rightarrow \int \cos^{2n-4} t dt = a^{2n-3} I_{n-1}$$

$$\Rightarrow I_n = \frac{1}{a^{2n-1}} \left[\frac{1}{2n-2} \left(\frac{a}{\sqrt{x^2 + a^2}} \right)^{2n-3} \cdot \frac{x}{\sqrt{x^2 + a^2}} + \frac{2n-3}{2n-2} a^{2n-3} I_{n-1} \right]$$

$$= \frac{1}{a^{2n-1}} \frac{a^{2n-3}}{2n-2} \left[\frac{x}{(x^2 + a^2)^{n-1}} + (2n-3) I_{n-1} \right]$$



II. Id.:

$$I_n = \int \frac{dx}{(x^2+a^2)^n} = \frac{1}{a^2} \int \frac{a^2 dx}{(x^2+a^2)^n} = \frac{1}{a^2} \int \frac{a^2 + x^2 - x^2}{(x^2+a^2)^n} dx = \frac{1}{a^2} \underbrace{\int \frac{dx}{(x^2+a^2)^{n-1}}}_{I_{n-1}} - \frac{1}{a^2} \underbrace{\int \frac{x^2}{(x^2+a^2)^n} dx}_A$$

$$I_n = \frac{1}{a^2} (I_{n-1} - A)$$

$$A = \int \frac{x^2 dx}{(x^2+a^2)^n} = \int x \cdot \frac{x dx}{(x^2+a^2)^n}$$

$$x^2+a^2 = t \Rightarrow 2x dx = dt \Rightarrow x dx = \frac{dt}{2}$$

$$\frac{x dx}{(x^2+a^2)^n} = dv \Rightarrow v = \int \frac{x dx}{(x^2+a^2)^n} = \frac{1}{2} \int \frac{dt}{t^n} = \frac{1}{2} \frac{t^{-n+1}}{-n+1} = -\frac{1}{2(n-1)t^{n-1}} \\ = -\frac{1}{(2n-2)(x^2+a^2)^{n-1}}$$

$$A = \int x \cdot \frac{x dx}{(x^2+a^2)^n} = \int x dv = xv - \int v dx = \frac{-x}{(2n-2)(x^2+a^2)^{n-1}} + \frac{1}{(2n-2)} \int \frac{dx}{(x^2+a^2)^{n-1}} \\ = \frac{-x}{(2n-2)(x^2+a^2)^{n-1}} + \frac{1}{2n-2} I_{n-1}$$

$$I_n = \frac{1}{a^2} \left(I_{n-1} + \frac{x}{(2n-2)(x^2+a^2)^{n-1}} - \frac{1}{2n-2} I_{n-1} \right)$$

$$= \frac{1}{a^2} \left[\frac{x}{(2n-2)(x^2+a^2)^{n-1}} - \frac{2n-3}{2n-2} I_{n-1} \right]$$

$$= \frac{1}{a^2(2n-2)} \left[\frac{x}{(x^2+a^2)^{n-1}} - (2n-3) I_{n-1} \right]$$

$$46) I_n = \int x^n e^x dx$$

$$n \geq 1, I_2 = ?$$

ind. II

$$x^n = u \Rightarrow n x^{n-1} dx = du \quad e^x dx = v \Rightarrow v = e^x$$

$$\int x^n e^x dx = uv - \int v du = x^n e^x - n \int x^{n-1} e^x dx = x^n e^x - n I_{n-1}$$

$$I_n = x^n e^x - n I_{n-1}$$

$$I_0 = e^x + c_1$$

$$I_1 = x e^x - 1 \cdot (e^x + c_1) = x e^x - e^x + c_2 \quad (c_2 = -c_1)$$

$$I_2 = x^2 e^x - 2 \cdot (x e^x - e^x + c_2) = x^2 e^x - 2 x e^x + 2 e^x + c_3 \quad (c_3 = -2 c_2) \\ = e^x (x^2 - 2x + 2) + c_3 //$$

$$45) I_n = \int (\ln x)^n dx$$

$$(\ln x)^n = u \Rightarrow n \cdot (\ln x)^{n-1} \cdot \frac{1}{x} dx = du$$

$$I_n = \int (\ln x)^n dx = \int u \cdot dx = x \cdot u - \int x du = x \cdot (\ln x)^n - n \int x (\ln x)^{n-1} \cdot \frac{1}{x} dx \\ = x \cdot (\ln x)^n - n \int (\ln x)^{n-1} dx$$

$$I_n = x \cdot (\ln x)^n - n \cdot I_{n-1}$$

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$$(46) I = \int x^3 \cos x^2 dx = \int x^2 \cdot x \cos x^2 dx$$

$$dv = x \cdot \cos x^2 dx \Rightarrow v = \frac{\sin x^2}{2} \quad u = x^2 \Rightarrow 2x dx = du$$

$$I = \int x^3 \cos x^2 dx = \int x^2 \cdot x \cos x^2 dx = \int u dv = uv - \int v \cdot du$$

$$= \frac{1}{2} x^2 \sin x^2 - \int x \sin x^2 dx = \frac{1}{2} x^2 \sin x^2 + \frac{1}{2} \cos x^2 + C$$

$$(47) I_n = \int \frac{dx}{(x^2+1)^n} = \int \frac{(x^2+1-x^2) dx}{(x^2+1)^n} = \underbrace{\int \frac{dx}{(x^2+1)^{n-1}}}_{I_{n-1}} - \underbrace{\int \frac{x^2 dx}{(x^2+1)^n}}_A$$

$$I_n = I_{n-1} - A$$

$$\frac{x dx}{(x^2+1)^n} = dv \Rightarrow v = \int \frac{x dx}{(x^2+1)^n} = \frac{1}{2} \int \frac{dt}{t^n} = \frac{1}{2} \frac{t^{-n+1}}{-n+1} = \frac{-1}{(2n-2)(x^2+1)^{n-1}}$$

$$x^2+1 = t \Rightarrow 2x dx = dt$$

$$A = \int x \cdot \frac{x dx}{(x^2+1)^n} = \int x dv = x \cdot v - \int v dx = \frac{-x}{(2n-2)(x^2+1)^{n-1}} + \frac{1}{(2n-2)} \int \frac{dx}{(x^2+1)^{n-1}}$$

$$= \frac{-x}{(2n-2)(x^2+1)^{n-1}} + \frac{1}{2n-2} I_{n-1}$$

$$I_n = I_{n-1} + \frac{x}{(2n-2)(x^2+1)^{n-1}} - \frac{1}{(2n-2)} I_{n-1}$$

$$I_n = \frac{1}{(2n-2)} \left\{ (2n-3) I_{n-1} + \frac{x}{(x^2+1)^{n-1}} \right\}$$

TRIGONOMETRIC INTEGRALS

$$\textcircled{1} \int \cos^3 x \, dx = \int \cos^2 x \cos x \, dx = \int (1 - \sin^2 x) \cos x \, dx$$

$$\sin x = t \Rightarrow \cos x \, dx = dt$$

$$\int \cos^3 x \, dx = \int (1 - t^2) \, dt = \int dt - \int t^2 \, dt = t - \frac{t^3}{3} + C = \sin x - \frac{1}{3} \sin^3 x + C$$

$$\textcircled{2} \int \sin^5 x \cos^2 x \, dx = \int (\sin^2 x)^2 \cdot \cos x \cdot \sin x \, dx = \int (1 - \cos^2 x)^2 \cos x \cdot \sin x \, dx$$

$$\cos x = t \Rightarrow -\sin x \, dx = dt$$

$$\int \sin^5 x \cos^2 x \, dx = \int (1 - t^2)^2 \cdot t^2 \cdot (-dt) = -\int (t^6 - 2t^4 + t^2) \, dt$$

$$= -\frac{t^7}{7} + \frac{2t^5}{5} - \frac{t^3}{3} + C =$$

$$= -\frac{1}{7} \cos^7 x + \frac{2}{5} \cos^5 x - \frac{1}{3} \cos^3 x + C$$

$$\textcircled{3} \int \sin^4 x \, dx = \int (\sin^2 x)^2 \, dx = \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx$$

$$= \frac{1}{4} \int (1 - 2\cos 2x + \cos^2 2x) \, dx$$

$$= \frac{1}{4} \left(x - \sin 2x + \int \cos^2 2x \, dx \right)$$

$$= \frac{1}{4} \left(x - \sin 2x + \int \frac{1 + \cos 4x}{2} \, dx \right)$$

$$= \frac{x}{4} - \frac{\sin 2x}{4} + \frac{x}{8} + \frac{\sin 4x}{32} + C$$

$$= \frac{3x}{8} + \frac{\sin 4x}{32} - \frac{\sin 2x}{4} + C$$

$$\tan^{2n} \sec^{2n} \Rightarrow \tan^{2n} = 1$$

$$\textcircled{4} \int \tan^6 x \sec^4 x dx = \int \tan^6 x \cdot \sec^2 x \cdot \sec^2 x dx = \int \tan^6 x (1 + \tan^2 x) \sec^2 x dx$$

$$\tan x = u \Rightarrow \sec^2 x dx = du$$

$$\begin{aligned} \int \tan^6 x \sec^4 x dx &= \int u^6 (1 + u^2) du = \int (u^6 + u^8) du = \frac{u^7}{7} + \frac{u^9}{9} + C \\ &= \frac{1}{9} \tan^9 x + \frac{1}{7} \tan^7 x + C \end{aligned}$$

$$\textcircled{5} \int \tan^5 \theta \sec^7 \theta d\theta = \int \tan^4 \theta \sec^6 \theta \cdot \tan \theta \sec \theta d\theta$$

$$\tan^2 \theta = \sec^2 \theta - 1 \quad \sec \theta = u \Rightarrow \sec \theta \tan \theta d\theta = du$$

$$\begin{aligned} \int \tan^5 \theta \sec^7 \theta d\theta &= \int (u^2 - 1)^2 \cdot u^6 du = \int (u^{10} - 2u^8 + u^6) du \\ &= \frac{1}{11} u^{11} - \frac{2}{9} u^9 + \frac{1}{7} u^7 + C = \frac{1}{11} \sec^{11} \theta - \frac{2}{9} \sec^9 \theta + \frac{1}{7} \sec^7 \theta + C \end{aligned}$$

$$\textcircled{6} \int \sec x dx = \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} dx$$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx$$

$$= \int \frac{du}{u} = \ln|u| + C = \ln|\sec x + \tan x| + C$$

$$\tan x + \sec x = u$$

$$(\sec^2 x + \sec x \tan x) dx = du$$

$$\textcircled{7} \int \sec^4 x dx = \int \sec^2 x \sec^2 x dx = \int (1 + \tan^2 x) \sec^2 x dx$$

$$= \int (1 + u^2) du = u + \frac{u^3}{3} + C = \tan x + \frac{1}{3} \tan^3 x + C$$

$$\tan x = u$$

$$\sec^2 x dx = du$$

$$\textcircled{8} \int \tan^4 x dx = \int (\tan^2 x)^2 dx = \int (\sec^2 x - 1)^2 dx$$

$$= \int \sec^4 x - 2 \int \sec^2 x + \int dx$$

$$= \tan x + \frac{1}{3} \tan^3 x - 2 \tan x + x + C = \frac{1}{3} \tan^3 x - \tan x + x + C$$

$$\textcircled{9} \int \tan^3 x \, dx = \int \tan^2 x \tan x \, dx = \int (\sec^2 x - 1) \tan x \, dx$$

$$= \int \tan x \sec^2 x \, dx - \int \tan x \, dx$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$= \int u \, du - \ln |\sec x|$$

$$= \frac{1}{2} \tan^2 x - \ln |\sec x| + C$$

$$\textcircled{10} \int \sec^3 x \, dx = \int \sec x \cdot \sec^2 x \, dx$$

$$\sec x = u \Rightarrow \sec x \tan x \, dx = du, \sec^2 x \, dx = dv \Rightarrow v = \tan x$$

$$\int \sec x \cdot \sec^2 x \, dx = \int u \, dv = uv - \int v \, du = \sec x \tan x - \int \tan^2 x \sec x \, dx$$

$$= \sec x \tan x - \int (\sec^2 x - 1) \sec x \, dx$$

$$= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

$$2 \int \sec^3 x \, dx = \sec x \tan x + \int \sec x \, dx = \sec x \tan x + \ln |\sec x + \tan x| + C,$$

$$\Rightarrow \int \sec^3 x \, dx = \frac{1}{2} [\sec x \tan x + \ln |\sec x + \tan x|] + C$$

$$\textcircled{11} \int \sin 4x \cdot \cos 5x \, dx = \frac{1}{2} \int [\sin(x) + \sin 9x] \, dx$$

$$= \frac{1}{2} \left[\cos x - \frac{1}{9} \cos 9x \right] + C$$

$$\textcircled{12} \int \frac{dx}{1 - \cos x} = \int \frac{1 + \cos x}{\sin^2 x} \, dx = \int \frac{dx}{\sin^2 x} + \int \frac{\cos x \, dx}{\sin^2 x} = -\cot x + \int \frac{du}{u^2}$$

$$= -\cot x - \frac{1}{u} + C = -\cot x - \frac{1}{\sin x} + C$$

$$(13) \int \frac{\tan^3 x}{\cos^3 x} dx = \int \tan^2 x \sec^3 x dx = \int \tan^2 x \sec^2 x \sec x \tan x dx$$

$$\sec x = u \Rightarrow \sec x \tan x dx = du$$

$$\begin{aligned} \int \frac{\tan^3 x}{\cos^3 x} dx &= \int (\sec^2 x - 1) \sec x \sec x \tan x dx = \int (u^2 - 1) u^2 du = \int u^4 du - \int u^2 du \\ &= \frac{1}{5} u^5 - \frac{1}{3} u^3 + C = \frac{1}{5} \sec^5 x - \frac{1}{3} \sec^3 x + C \end{aligned}$$

II. Vol.:

$$\int \frac{\tan^3 x}{\cos^2 x} dx = \int \frac{\sin^2 x dx}{\cos^6 x} = \int \frac{(1 - \cos^2 x) \sin x dx}{\cos^6 x}$$

$$\cos x = u$$

$$-\sin x dx = du$$

$$= \int \frac{u^2 - 1}{u^6} du = \int \frac{-1}{u^6} du + \int \frac{1}{u^4} du$$

$$= \frac{1}{5} u^{-5} - \frac{1}{3} u^{-3} + C = \frac{1}{5} \sec^5 x + \frac{1}{3} \sec^3 x + C$$

$$(14) \int \frac{\sqrt{1-x}}{\sqrt{1+x}} dx = \int \frac{1-x}{\sqrt{1-x^2}} dx = \int \frac{dx}{\sqrt{1-x^2}} - \int \frac{x}{\sqrt{1-x^2}} dx$$

$$1-x^2 = u$$

$$x dx = -\frac{1}{2} du$$

$$= \arcsin x + \frac{1}{2} \int \frac{du}{\sqrt{u}} = \arcsin x + \sqrt{u} + C$$

$$= \arcsin x + \sqrt{1-x^2} + C$$

$$(15) \int \sec^5 x dx = \int \sec^3 x \sec^2 x dx$$

$$\sec^2 x = u \Rightarrow 3 \sec^2 x \sec x \tan x dx = du \quad \sec^2 x dx = dv \Rightarrow v = \tan x$$

$$\int u dv = uv - \int v du = \sec^3 x \tan x - 3 \int \sec^3 x \tan^2 x dx$$

$$= \sec^3 x \tan x - 3 \int \sec^3 x (\sec^2 x - 1) dx = \sec^3 x \tan x - 3 \int \sec^5 x dx + 3 \int \sec^3 x dx$$

$$\Rightarrow 4 \int \sec^5 x dx = \sec^3 x \tan x + 3 \cdot \frac{1}{2} [\sec x \tan x + \ln |\sec x + \tan x|] + C$$

$$\int \sec^5 x dx = \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \ln |\sec x + \tan x| + C$$

$$(16) \int \frac{dx}{1+\sin x}$$

$$\tan \frac{x}{2} = t \Rightarrow dx = \frac{2dt}{1+t^2} \quad \sin x = \frac{2t}{1+t^2} \Rightarrow 1+\sin x = \frac{1+2t+t^2}{1+t^2}$$

$$I = \int \frac{dx}{1+\sin x} = \int \frac{2dt/1+t^2}{\frac{1+2t+t^2}{1+t^2}} = \int \frac{2dt}{(1+t)^2} = \frac{-2}{1+t} + C = \frac{-2}{1+\tan \frac{x}{2}} + C$$

$$(17) \int \frac{dx}{3+\cos x+2\sin x}$$

$$\tan \frac{x}{2} = t \Rightarrow dx = \frac{2dt}{1+t^2} \quad \sin x = \frac{2t}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$$

$$\begin{aligned} \int \frac{dx}{3+\cos x+2\sin x} &= \int \frac{2dt/1+t^2}{3+\frac{1-t^2}{1+t^2}+\frac{4t}{1+t^2}} = \int \frac{2dt}{3+3t^2+1-t^2+4t} \\ &= \int \frac{2dt}{2t^2+4t+4} = \int \frac{dt}{t^2+2t+2} = \int \frac{dt}{(t+1)^2+1} \quad \begin{matrix} t+1=s \\ dt=ds \end{matrix} \\ &= \int \frac{ds}{1+s^2} = \arctan s + C = \arctan(1+t) + C \\ &= \arctan\left(1+\tan \frac{x}{2}\right) + C \end{aligned}$$

$$(18) \int \sin^3 x \cos^8 x dx = \int \sin^2 x \cos^8 x \sin x dx = \int (1-\cos^2 x) \cos^8 x \sin x dx$$

$$\cos x = u \Rightarrow \sin x dx = -du$$

$$\begin{aligned} \int \sin^2 x \cos^8 x dx &= \int (1-u^2) \cdot u^8 \cdot (-du) = \int (u^8 - u^{10}) du = \frac{1}{11} u^{11} - \frac{1}{9} u^9 + C \\ &= \frac{1}{11} \cos^{11} x - \frac{1}{9} \cos^9 x + C \end{aligned}$$

$$(19) \int \cos^5 ax dx = \int \cos^4 ax \cdot \cos ax dx = \int (1-\sin^2 ax)^2 \cos ax dx$$

$$\sin ax = u \Rightarrow \cos ax dx = \frac{1}{a} du$$

$$\begin{aligned} \int \cos^5 ax dx &= \frac{1}{a} \int (1-u^2)^2 du = \frac{1}{a} \int (1-2u^2+u^4) du = \frac{1}{a} \left(u - \frac{2}{3} u^3 + \frac{1}{5} u^5 \right) + C \\ &= \frac{1}{a} \left(\sin ax - \frac{2}{3} \sin^3 ax + \frac{1}{5} \sin^5 ax \right) + C \end{aligned}$$

$$(20) \int \sin^3 x \cdot \sqrt{\cos x} \, dx = \int (1 - \cos^2 x) \sqrt{\cos x} \cdot \sin x \, dx$$

$$\cos x = u \quad \sin x \, dx = -du$$

$$\begin{aligned} \int \sin^3 x \sqrt{\cos x} \, dx &= \int (1 - u^2) u^{1/2} \, du = \int (u^{3/2} - u^{5/2}) \, du = \frac{u^{7/2}}{7/2} - \frac{u^{3/2}}{3/2} + C \\ &= \frac{2}{7} \sqrt{\cos^7 x} - \frac{2}{3} \sqrt{\cos^3 x} + C \end{aligned}$$

$$(21) \int \operatorname{cosec}^4 x \cdot \cot^6 x \, dx = \int \cot^6 x \cdot \operatorname{cosec}^2 x \cdot \operatorname{cosec}^2 x \, dx$$

$$\cot x = u \Rightarrow \operatorname{cosec}^2 x \, dx = -du \quad \operatorname{cosec}^2 x = 1 + \cot^2 x = 1 + u^2$$

$$\begin{aligned} \int \operatorname{cosec}^4 x \cdot \cot^6 x \, dx &= \int u^6 \cdot (1 + u^2) (-du) = \int (-u^8 - u^6) \, du = -\frac{1}{9} u^9 - \frac{1}{7} u^7 + C \\ &= -\frac{1}{9} \cot^9 x - \frac{1}{7} \cot^7 x + C \end{aligned}$$

$$(22) \int \sin 5x \cdot \sin 2x \, dx = \int \frac{1}{2} (\sin 7x + \sin 3x) \, dx$$

$$= -\frac{1}{14} \cos 7x - \frac{1}{6} \cos 3x + C$$

$$(23) \int \cos 7\theta \cdot \cos 5\theta \, d\theta = \int \frac{1}{2} (\cos 12\theta + \cos 2\theta) \, d\theta$$

$$= \frac{1}{24} \sin 12\theta + \frac{1}{4} \sin 2\theta + C$$

$$(24) \int \tan^5 x \sec^4 x \, dx$$

$$\tan^2 x = u \Rightarrow 2 \tan x \cdot \sec^2 x \, dx = du \Rightarrow \tan x \sec^2 x \, dx = \frac{du}{2}$$

$$\int \tan^5 x \sec^4 x \, dx = \int \tan^4 x \sec^2 x \cdot \tan x \sec^2 x \, dx$$

$$= \int u^2 \cdot (1 + u) \, du = \int (u^3 + u^2) \, du = \frac{1}{4} u^4 + \frac{1}{3} u^3 + C$$

$$= \frac{1}{4} \tan^4 x + \frac{1}{3} \tan^3 x + C$$

$$(25) \int \tan^5 x \cdot \sec^6 x \, dx$$

$$\tan^2 x = u \Rightarrow \tan x \sec^2 x \, dx = \frac{du}{2} \quad \sec^2 x = 1 + u$$

$$\int \tan^5 x \sec^6 x \, dx = \int \tan^4 x \cdot \sec^4 x \tan x \sec^2 x \, dx = \int u^2 (1+u)^2 \, du$$

$$= \int (u^6 + 2u^4 + u^2) \, du = \frac{1}{7} u^7 + \frac{2}{5} u^5 + \frac{1}{3} u^3 + C$$

$$= \frac{1}{7} \tan^7 x + \frac{2}{5} \tan^5 x + \frac{1}{3} \tan^3 x + C //$$